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SynchLend AI: Enterprise Credit Decisioning for NTC Borrowers

Multi-Modal Vector RAG Underwriting Engine

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THE CHALLENGE

The "New-To-Credit" (NTC) Blindspot

Millions of creditworthy applicants — gig workers, students, and cash-flow earners — are invisible to traditional underwriting. Legacy systems were never designed to assess them.

Bureau Coverage Gap

Traditional credit bureaus fail to evaluate applicants without established credit histories, leaving millions of financially active individuals without a credit score.

Manual Underwriting at Scale

Evaluating alternative data — gig-economy income, unstructured cash flow — is slow, expensive, and prone to human bias, creating operational bottlenecks.

Black-Box AI Risk

Opaque AI models generate lending decisions without regulatory transparency or policy justification, creating compliance exposure for enterprise lenders.

The Goal: Build an automated, transparent, and mathematically grounded underwriting engine that leverages unstructured data safely.



THE SOLUTION

SynchLend AI: A Next-Generation Underwriting Architecture

SynchLend AI replaces opaque, manual underwriting with a structured, policy-aware AI engine — combining semantic retrieval, mathematical guardrails, and human-readable explanations in a single decision pipeline.



Dynamic Feature Extraction

Translates unstructured employment and cash flow notes into quantifiable stability metrics that feed directly into risk scoring.



Semantic RAG Engine

Queries corporate lending policies instantly to ensure every decision aligns with regulatory thresholds and institutional rules.



Transparent AI

Generates human-readable, policy-backed explanations for every approval or decline — eliminating the AI "black box" entirely.



Mathematical Guardrails

Hardcoded Debt-to-Income (DTI) and risk score floors override AI hallucinations, ensuring deterministic compliance.

SYSTEM DESIGN

End-to-End Enterprise Architecture

SynchLend AI is built on a modern, cloud-ready stack designed for enterprise-grade performance, modularity, and seamless scalability from local development to AWS production.



Frontend & Backend

A **React JS** interface (Vite) with real-time state management connects to a **FastAPI** backend — a high-performance, asynchronous API service engineered for low-latency credit decisioning.

Data & AI Layer

SQLite handles local operational logging, with a migration path to **PostgreSQL + pgvector** for cloud-scale structured data and semantic search. The AI layer pairs **Google Gemini 1.5 Flash** with native embeddings for RAG-powered policy retrieval.

AI CORE

Advanced Retrieval-Augmented Generation

The RAG layer is the intelligence core of SynchLend AI — transforming raw applicant narratives into structured, policy-grounded credit decisions with deterministic fallbacks.



Embedding Model

Transforms unstructured applicant notes — employment descriptions, cash flow narratives — into high-dimensional vectors for semantic comparison against policy documents.



Semantic Policy Search

ChromaDB dynamically retrieves the most relevant corporate underwriting rule — distinguishing gig-worker guidelines from corporate employee thresholds in milliseconds.



Prompt Templates & Guardrails

Strict contextual prompts force the LLM to synthesize decisions based *only* on retrieved mathematical data and policy — preventing hallucinated or off-policy outputs.



Deterministic Fallbacks

If the LLM API fails or returns an ambiguous response, the system auto-routes to a hardcoded mathematical explanation pipeline, ensuring zero decision gaps.

ENGINEERING

Production-Ready Code & Security

SynchLend AI is engineered with enterprise security discipline from the ground up — every layer enforces strict boundaries, isolation, and auditability to meet institutional compliance standards.



Input Validation

Strict **Pydantic models** enforce payload boundaries — limiting income string lengths, numeric ranges, and field types — to prevent injection attacks and malformed data entry.



No Hardcoded Secrets

All API keys and credentials are cryptographically isolated using **.env variables** and **.gitignore**, ensuring secrets never enter version control or production logs.



Secure API Usage

CORS middleware is restricted to designated frontend origins only, preventing unauthorized cross-origin requests and protecting the backend from external exploitation.



API-First Design

Clean, modular separation of **routing, scoring, vector storage,** and **database persistence** enables independent testing, deployment, and scaling of each subsystem.

QA & CLOUD

Testing & Deployment Readiness

Automated Test Coverage

An automated **pytest suite** validates unit logic across diverse real-world scenarios — including High DTI rejections, Negative Cash Flow edge cases, and Optimal Approval paths — ensuring consistent behavior before any deployment.

Version Control

Full project managed via **Git/GitHub** with clear README setup instructions, enabling seamless onboarding for enterprise engineering teams.

Cloud Architecture

Designed for **AWS deployment** utilizing managed **PostgreSQL (RDS)** for structured data persistence and **AWS Secrets Manager** for secure credential rotation and access control.

Audit Logging

Every transaction, RAG match, and AI output is **timestamped and committed** to the database — creating a tamper-evident compliance trail for regulatory monitoring and audit reviews.

RESPONSIBLE AI

Eliminating the AI Black Box

Regulatory bodies and enterprise risk teams demand explainability. SynchLend AI delivers full transparency by separating mathematical scoring, policy retrieval, and AI rationale into distinct, auditable layers.

Total Transparency

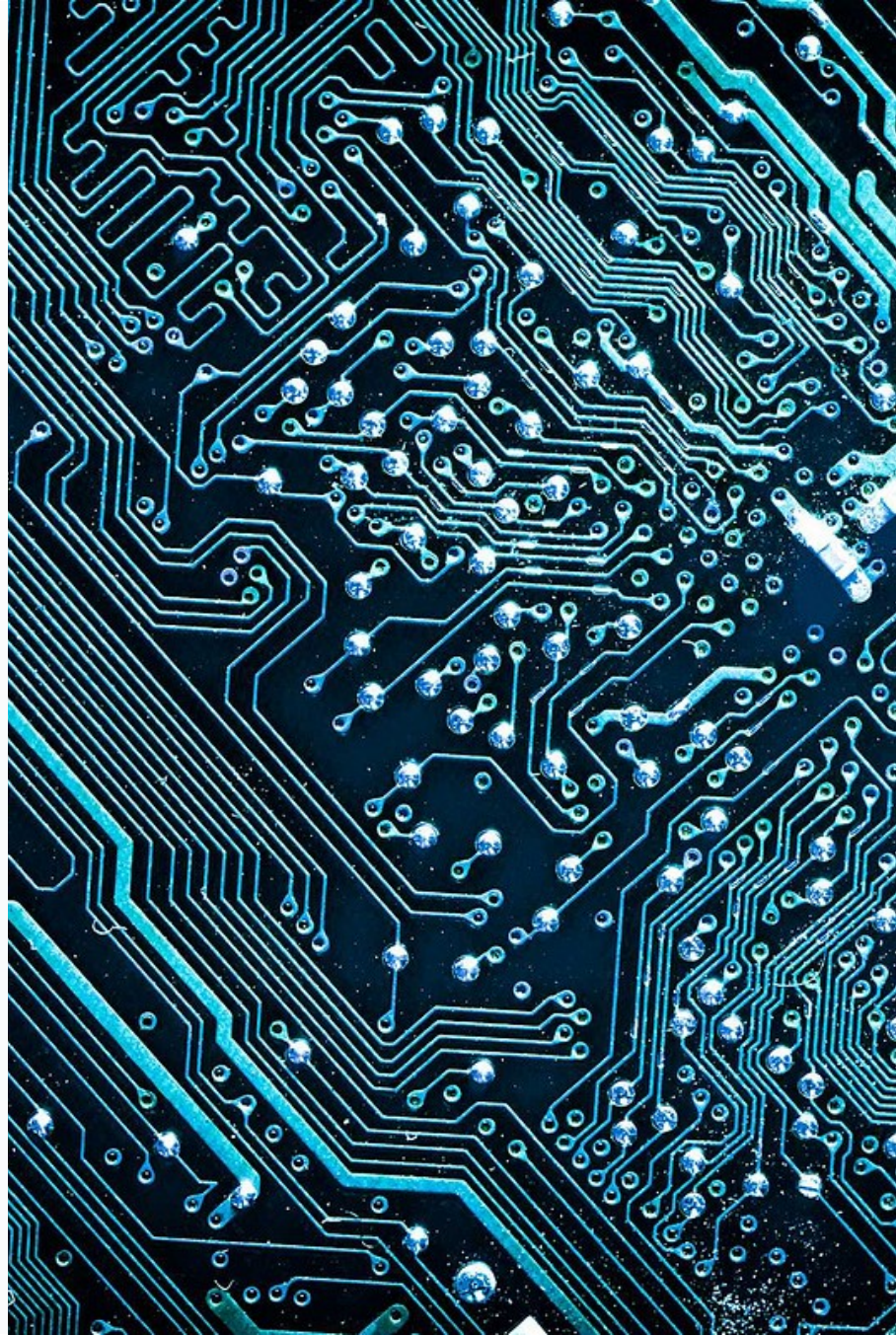
The UI explicitly splits three components: the **mathematical Risk Score**, the **retrieved Policy Guideline**, and the **AI's natural language rationale** — each independently verifiable.

Anti-Parroting Protocol

The LLM is restricted from blindly copying policy text. It must **synthesize the applicant's exact numbers** against the rule, producing a contextualized, case-specific explanation.

Bias-Free Fairness

Sentiment analysis scores alternative data objectively, **removing subjective human loan-officer bias** from the decision process and ensuring consistent treatment across applicant demographics.

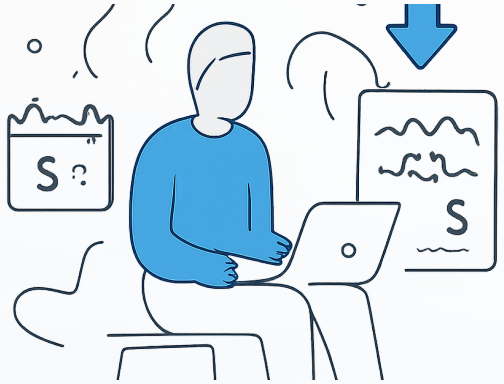


LIVE DEMO

Live System Walkthrough

The following scenarios demonstrate SynchLend AI's full decision pipeline — from raw applicant data through RAG policy retrieval to a transparent, explainable credit decision.





Scenario 1: Tier 1 Corporate Approval

Demonstrates **DTI math and stable sentiment scoring** for a salaried corporate applicant. The system calculates a clean risk score, retrieves standard employment policy, and generates a concise approval rationale backed by hard numbers.

Scenario 2: Gig-Economy Policy Rejection

Demonstrates **Vector RAG dynamically pulling independent contractor guidelines**. ChromaDB retrieves the gig-worker policy, the LLM synthesizes the applicant's irregular cash flow against stricter thresholds, and a policy-backed decline is issued.

A recorded video demo is included in the submission ZIP package, covering both scenarios end-to-end with full UI walkthrough and backend log inspection.